

Desert Detectives

How do animals survive in the desert?

Grade: 2 | **Content:** Science + ENL

Proficiency: Emerging-Transitioning | **Duration:** 60 minutes

ENGLISH + SPANISH EDITION | Paired student materials on pages 5-7. Use these labels, prompts, and frames during the lesson sequence.

Portfolio focus: Visual scaffolding, vocabulary development, oral language, sentence frames, and collaborative learning.

Lesson context

Students are learning about habitats and animal survival. This sample scenario centers multilingual learners who benefit from photographs, modeled language, repeated vocabulary exposure, partner interaction, and home-language connections. It describes planned instruction, not documented student outcomes.

Objectives

Content: Identify at least two characteristics of a desert habitat and explain one way a desert animal survives there.

Language: Describe an animal and explain a survival feature orally and/or in writing using a complete supported sentence.

A ____ survives in the desert by ____.

Because the desert is ____, the ____ uses ____ to survive.

Key vocabulary

desert, habitat, survive, adaptation, hot, dry, water, shade, nocturnal, camel, lizard, fennec fox. Teach **habitat**, **adaptation**, and **survive** explicitly; use animal names and other terms as picture-supported vocabulary.

Materials and preparation

Prepare a desert photograph; animal photographs; picture vocabulary cards; chart paper; markers; an animal organizer; short illustrated animal fact cards; and an exit ticket. Choose reliable animal facts and readable images before teaching.

Activate knowledge of familiar habitats. Clarify that deserts receive little precipitation; not every desert is hot. This lesson focuses on animals in hot deserts.

Observe. Describe. Explain.

1. See-Think-Wonder | 8 minutes

Display a desert photograph. Ask: What do you see? What do you think? What do you wonder? Give partners time to rehearse in English or a home language, then record ideas under SEE / THINK / WONDER.

Frames: **I see _____. I think _____. I wonder _____.** Recast responses without interrupting meaning. For example: “Yes, the desert is very dry and receives very little rain.”

2. Vocabulary instruction | 8 minutes

Introduce words with photographs, brief definitions, and gestures. A habitat is the place where an animal or plant lives. An adaptation is a body feature or behavior that helps an organism survive in its environment.

Students hear, see, gesture, match, and say the words. Check understanding through picture choices before asking for independent definitions.

3. Teacher modeling | 7 minutes

Display a camel. Explain that its hump stores **fat, not water**, which provides energy when food is scarce. Think aloud as you connect the animal, its feature, and the benefit.

Model: **A camel stores fat in its hump, which provides energy when food is scarce.**
Complete one organizer together.

4. Guided practice | 7 minutes

Display a fennec fox and ask what students notice. Explain that its large ears help release body heat. Build and rehearse: **A fennec fox survives in the desert because its large ears help it stay cool.**

5. Desert Detective Cards | 12 minutes

Pairs read one illustrated animal card and complete an organizer. Possible animals: camel, fennec fox, desert lizard, desert snake, or scorpion. Use species-specific facts rather than assuming all snakes or lizards behave alike.

Organizer prompts: **My animal / I notice / The desert is / This animal survives by.** Students may draw first, discuss in a home language, and then rehearse an English explanation.

6. Survival poster and exit ticket | 18 minutes

Allow 14 minutes for a mini poster and brief sharing, then 4 minutes for the exit ticket. Poster requirements: animal name, image or drawing, one desert characteristic, one survival feature or behavior, and an explanatory sentence.

Make the thinking visible

Differentiation

Entering/Emerging: Labeled pictures, small word banks, bilingual resources, gestures, partner rehearsal, and oral or dictated responses. Frames: “A ____ lives in the desert.” “It uses ____ to survive.”

Transitioning: Combine ideas using “The ____ has ____, which helps it ____.” Gradually remove the frame as students become more independent.

Expanding extension: Explain cause and effect using “Because the desert is ____, the ____ survives by ____.” Compare two animals using evidence from the cards.

Culturally and linguistically responsive practice

Invite students to connect the habitat to environments they know through family, books, travel, or community knowledge. Do not assume personal familiarity based on a student’s background.

Allow home-language discussion and bilingual vocabulary notes. Students may explain a familiar animal or choose one from the prepared cards.

Formative assessment

Listen to partner explanations; check picture-word matches, organizer entries, and use of target vocabulary. Ask: “How does that feature help the animal?” Provide another model if a student names a feature without explaining its function.

Culminating task: Desert Survival Poster

Check for: **(1)** accurate animal identification; **(2)** a relevant desert characteristic; **(3)** an accurate survival feature; **(4)** an explanation connecting the feature to survival. Record science understanding separately from the amount of language support used.

Example: “The desert is hot and dry. A fennec fox has large ears that help release heat and keep its body cool.”

Exit ticket

Complete: **One way a desert animal survives is ____.** Accept writing, a labeled drawing with an oral explanation, or a recorded response. Use responses to group students for reteaching or extension.

Accessibility

Provide enlarged, high-contrast images; read fact cards aloud; allow pointing, drawing, dictation, or speech-to-text where appropriate. Follow individual IEP/504 supports and provide additional processing time.

Why I designed it this way

The instructional challenge

Students must understand a science relationship while learning the English needed to explain it. This lesson keeps the scientific thinking intact and provides visual and linguistic support for participation.

The design decisions

Observation before terminology: See-Think-Wonder gives students an entry point through what they notice before requiring specialized vocabulary.

Modeling before independent production: A think-aloud shows how to connect an animal's feature with its survival benefit. Guided practice makes that relationship explicit.

Oral rehearsal before writing: Partner talk lets students test an explanation and receive feedback before creating the poster.

Home language as a resource: Multilingual brainstorming supports meaning-making and helps students connect existing knowledge to new English language.

Evidence to collect after teaching

Retain an anonymized organizer, poster, and brief observation notes. Compare the initial description with the exit explanation, noting both scientific accuracy and independence with language. Obtain appropriate permission before displaying student work.

Reflection prompts

Which visual or sentence frame supported the clearest explanations? Could students explain why a feature helps rather than merely naming it? Who needs a new model, and who is ready to compare animals?

Design rationale only: This sample does not claim implementation, measured learning gains, or verified standards alignment.

Words for our learning

Palabras para aprender

Teacher use: Display paired labels beside the same image in equally readable type. Read and model both labels when possible. Invite Spanish or bilingual planning, then support English practice toward the language objective. Offer other language supports as needed; Spanish is not every learner's home language.

ENGLISH	ESPAÑOL
desert	desierto
habitat	hábitat
survive	sobrevivir
adaptation	adaptación
hot / dry	caluroso / seco
water / shade	agua / sombra
nocturnal: active at night	nocturno: activo durante la noche
camel / hump	camello / joroba
fennec fox / large ears	fénec (zorro del desierto) / orejas grandes
lizard / snake / scorpion	lagartija / serpiente / escorpión

Learning goals / Metas de aprendizaje

ENGLISH	ESPAÑOL
I can name two features of a desert and explain how one animal survives.	Puedo nombrar dos características de un desierto y explicar cómo sobrevive un animal.
I can use words and pictures to explain my idea.	Puedo usar palabras e imágenes para explicar mi idea.

Look, label, and explain

Observa, nombra y explica

Teacher preparation: These paired captions are ready to place beneath the selected photographs. The photographs themselves are not included.

ENGLISH	ESPAÑOL
Desert A desert receives very little rain.	Desierto Un desierto recibe muy poca lluvia.
Camel / Hump A camel stores fat in its hump. The fat provides energy when food is scarce.	Camello / Joroba El camello almacena grasa en su joroba. La grasa le proporciona energía cuando hay poca comida.
Fennec fox / Large ears Its large ears help release body heat.	Fénec / Orejas grandes Sus orejas grandes ayudan a liberar el calor del cuerpo.

See-Think-Wonder / Veo-Pienso-Me pregunto

ENGLISH	ESPAÑOL
What do you see? I see _____.	¿Qué ves? Veo _____.
What do you think? I think _____.	¿Qué piensas? Pienso que _____.
What do you wonder? I wonder _____.	¿Qué te preguntas? Me pregunto _____.

Explain with a partner / Explica con un compañero

ENGLISH	ESPAÑOL
The _____ has _____, which helps it _____.	El/La _____ tiene _____, lo que le ayuda a _____.
Because the desert is _____, the animal _____.	Como el desierto es _____, el animal _____.

My desert animal

Mi animal del desierto

ENGLISH	ESPAÑOL
Read the animal card with a partner. Draw or write your ideas. You may plan in English, Spanish, or both.	Lee la tarjeta del animal con un compañero. Dibuja o escribe tus ideas. Puedes planear en inglés, español o ambos idiomas.
Animal: _____	Animal: _____
I notice: _____ _____	Observo: _____ _____
The desert is: _____ _____	El desierto es: _____ _____
This animal survives by: _____ _____	Este animal sobrevive al: _____ _____

Survival poster / Póster de supervivencia

ENGLISH	ESPAÑOL
Include the animal's name and a drawing or image. Label the image in English and Spanish.	Incluye el nombre del animal y un dibujo o una imagen. Escribe las etiquetas de la imagen en inglés y español.
Add one desert feature, one adaptation, and a sentence explaining how it helps the animal.	Añade una característica del desierto, una adaptación y una oración que explique cómo ayuda al animal.

Exit ticket / Boleto de salida

ENGLISH	ESPAÑOL
One way a desert animal survives is _____. _____	Una manera en que un animal sobrevive en el desierto es _____. _____
Write, draw and explain, or record your response.	Escribe, dibuja y explica, o graba tu respuesta.